



Build Pixel's Script

Write Pixel's code so they meet

Name: _____

Date: _____

★ I can write a script for a second sprite so both sprites meet on the same number.

Your turn to write code

Bit (green) already has a script. Pixel (purple) does NOT — that blank block is yours to fill. When the green flag ► starts BOTH at the same time, you want them to MEET (land on the SAME number). First find where Bit lands, then write Pixel's move so Pixel lands there too.

The number path



Bit

Bit's script (given)

► start at 0

forward 3

1 Where does Bit land?

Read Bit's script from 0: forward 3. What number does Bit land on?



Pixel

Pixel's script (you write the move)

▶ start at 5



2 Write Pixel's move

Pixel starts at 5. Write a move (forward or back, and how many) so Pixel lands on the SAME number as Bit. Pixel's move: _____

3 Check — do they meet?

The green flag starts BOTH scripts at once. Write where each lands now. Bit: ____ Pixel: _____. Do they meet? Circle YES or NO.

Big idea

Writing code means choosing the moves that reach your goal. To make two sprites meet, first find one sprite's number, then write the other sprite's script to land on that same number.



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Quick facilitation notes & answer key

What this worksheet teaches

Students WRITE moves for the first time in this subject: Bit's moves are given, and they write Pixel's move so the two characters MEET when the green flag starts both. Read Bit's result first, then write a move that matches it. No coding experience needed.

Before you start (no computer needed)

No computer needed — paper and a pencil. You read the prompts aloud. Optional: put two counters on a number line 0 to 5 — green for Bit, purple for Pixel — and move BOTH at the same time so children see the two sets of moves run side by side.

How to lead it (about 20 minutes)

1. Show them (you do). Find where Bit lands first, then ask: "what move puts Pixel on that same number?"
2. Do one together. Exercise 1 — Bit lands on 3.
3. Let them try. Students write Pixel's move to also reach 3 (Ex 2), then check that they meet (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 — Bit: 0, forward 3 → 3.

Exercise 2 — Pixel starts at 5 and must reach 3, so the move is back 2. (Any move that lands Pixel on 3 is correct; back 2 is simplest.)

Exercise 3 — with Pixel back 2, the green flag sends both to 3 → same number → YES, they meet.

If students get stuck — and ways to adjust

Watch for: writing a move without checking Bit's number first. Find Bit's landing, then aim Pixel at it.

Make it easier: count the steps from 5 to 3 on the line, then write that as a back move.

Make it harder: ask for a different Pixel move (two steps) that also reaches 3.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

In plain terms: students write and run instructions, including two sets that run at the same time.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

In plain terms: students read two sets of instructions, change one, and explain what happens when both run together.

You'll know it worked when

The child writes a Pixel move that lands on Bit's number so the two meet.